

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

/ P.Harsha vardhan(192525464) (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P.HARSHA VARDHAN

Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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ASSESSMENT TOOL 1 – ANNEXURE A: APPLICATION-BASED QUESTIONS

Course Name: Cryptography and Network Security Course Code: CSA51

CO1: Classical encryption using symmetric and asymmetric ciphers (BL3, BL4)

1. Event Communication Security

a. Encrypted message (Caesar cipher, shift = 4)

A student organization wants to share confidential event information with its core members. To protect the message, a simple classical encryption technique based on a **fixed shift pattern** is used. The given message is “**MEET AT NOON**”, and a shift of **4 positions** is applied.

a. Reconstruct the encrypted message using a shift of 4 positions

The encryption technique used here is the **Caesar cipher**. In this method, every letter in the plaintext is shifted by a fixed number of positions in the alphabet.

Step-by-Step Encryption:

1. The first letter M is shifted four positions forward and becomes Q.
2. The second letter E is shifted four positions forward and becomes I.
3. The third letter E is also shifted four positions forward and becomes I.
4. The fourth letter T becomes X.
5. A space is maintained between words.
6. The letter A becomes E.
7. The letter T becomes X.
8. The letter N becomes R.
9. The letter O becomes S.
10. The remaining letters are encrypted using the same shifting rule.

Mathematical Representation:

The Caesar Cipher encryption can be represented using the following

formula:

$$E(x)=(x+4)\text{mod}26$$

Where:

- E(x) represents the encrypted letter.
- x represents the numerical position of the original letter.
- 4 represents the fixed shift value.
- Modulo 26 ensures that the alphabet repeats after Z.

Here, the shift value is 4.

The alphabet is:

Plain alphabet : A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Shifted alphabet: E F G H I J K L M N O P Q R S T U V W X Y Z A B C D

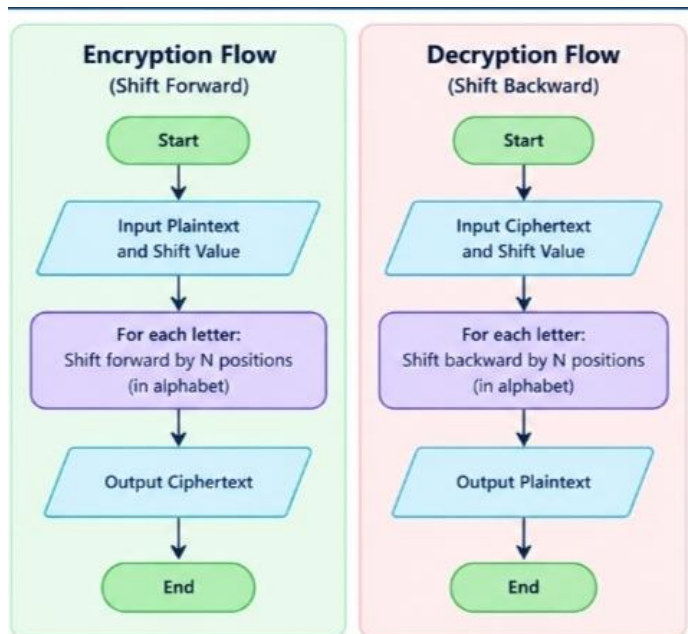
The scenario uses a fixed shift, so the Caesar cipher is applied. Each letter is replaced by the letter 4 positions ahead in the alphabet (A = 0, B = 1, ... Z = 25), wrapping around after Z. Encryption formula: $C = (P + 4) \text{ mod } 26$.

Plaintext	M	E	E	T	A	T	N	O	O	N
Position (P)	12	4	4	19	0	19	13	14	14	13
P + 4	16	8	8	23	4	23	17	18	18	17
Ciphertext	Q	I	I	X	E	X	R	S	S	R

Encrypted message: QIIX EX RSSR

b. Retrieval of the original message

An authorized member who knows the rule (Caesar cipher with key = 4) simply reverses the process by shifting every letter 4 positions backward. Decryption formula: $P = (C - 4) \bmod 26$.



- $Q (16) - 4 = 12 \rightarrow M$; $I (8) - 4 = 4 \rightarrow E$; $I \rightarrow E$; $X (23) - 4 = 19 \rightarrow T$. So $QIIX \rightarrow MEET$.
- $E (4) - 4 = 0 \rightarrow A$; $X (23) - 4 = 19 \rightarrow T$. So $EX \rightarrow AT$.
- $R (17) - 4 = 13 \rightarrow N$; $S (18) - 4 = 14 \rightarrow O$; $S \rightarrow O$; $R \rightarrow N$. So $RSSR \rightarrow NOON$.
-

Recovered message: MEET AT NOON. Since the same key is used for encryption and decryption, this is a symmetric technique; the secrecy depends entirely on keeping the shift value private.

Conclusion:

The Caesar cipher is a simple symmetric classical encryption technique because the same shift value is used as the secret rule for encryption and its reverse for decryption. In this example, the shift value is **4**. Although it is easy to implement and understand, a fixed-shift cipher provides only limited security because there are only a small number of possible shifts that an attacker can test.

3. Secure Messaging System

a. Step-by-step encryption (Vigenère cipher)

A keyword-based polyalphabetic cipher (Vigenère) is used. Plaintext = HELLO, Keyword = KEY. Encryption formula: $C = (P + K) \bmod 26$, with A = 0 ... Z = 25.

Step 1 – Repeat the key to match the message length: K E Y K E

Step 2 – Convert letters to numbers and add:

Plaintext	H	E	L	L	O
P value	7	4	11	11	14
Key letter	K	E	Y	K	E
K value	10	4	24	10	4
P + K	17	8	35	21	18
(P + K) mod 26	17	8	9	21	18
Ciphertext	R	I	J	V	S

Step 4: Perform Encryption

The Vigenère Cipher uses the following encryption formula:

$$C_i = (P_i + K_i) \bmod 26$$

Where:

- C represents the ciphertext value.
- P represents the plaintext value.
- K represents the keyword value.
- mod 26 ensures that the result remains within the alphabet range.

Plaintext	Value	Key	Key Value	Calculation	Ciphertext
H	7	K	10	$(7+10) \bmod 26 = 17$	R
E	4	E	4	$(4+4) \bmod 26 = 8$	I
L	11	Y	24	$(11+24) \bmod 26 = 9$	J
L	11	K	10	$(11+10) \bmod 26 = 21$	V
O	14	E	4	$(14+4) \bmod 26 = 18$	S

Ciphertext: RIJVS (Step 3 – convert the resulting numbers back to letters).
Decryption

reverses this with $P = (C - K) \bmod 26$.

Detailed Character Encryption:

- First character: $H (7) + K (10) = 17$, which represents R.
- Second character: $E (4) + E (4) = 8$, which represents I.
- Third character: $L (11) + Y (24) = 35$. After applying modulo 26, the result is 9, which represents J.
- Fourth character: $L (11) + K (10) = 21$, which represents V.
- Fifth character: $O (14) + E (4) = 18$, which represents S.

b. Influence of the repeating keyword

- Each plaintext letter is shifted by the amount of the key letter aligned with it: H is shifted by 10 (K), E by 4 (E), L by 24 (Y), and so on. The key therefore acts as a sequence of different Caesar shifts.
- The keyword is repeated cyclically (period = 3). The 4th letter reuses K and the 5th reuses E, so the shift pattern 10-4-24 repeats every three characters.
- The same plaintext letter can produce different ciphertext letters: the two L's in HELLO become J (shift Y) and V (shift K). This hides letter frequencies and defeats simple frequency analysis.
- Conversely, different plaintext letters can map to the same ciphertext letter, so the substitution is not one-to-one across the message. The repetition of the key is still the weakness: for long messages, Kasiski examination can reveal the key length.

c. Applications of Vigenère Cipher:

4. Secure Message Demonstrations: It demonstrates how keywords can control encryption and decryption processes.
5. Educational Purposes: It is widely used to explain classical cryptographic concepts.
6. Historical Communication: It was historically used to protect written messages and military communications.
7. Cryptographic Algorithm Learning: It helps students understand polyalphabetic substitution and key-based encryption.
8. Understanding Modern Security: It provides a foundation for understanding the importance of secret keys in modern encryption systems.

Conclusion:

The Vigenère Cipher is a classical polyalphabetic substitution technique that uses a repeating keyword to encrypt and decrypt messages. In this scenario, the company encrypts the plaintext "HELLO" using the keyword "KEY", producing the ciphertext "RIJVS". The repeating keyword KEYKE determines the individual shifting values for each character, making the encryption process different from a simple fixed-shift Caesar Cipher. The receiver can retrieve the original message by applying the reverse operation using the same keyword. This technique demonstrates the importance of secret keys in cryptography and provides a foundation for understanding more advanced encryption methods. However, because it uses a repeating keyword, it is vulnerable to modern cryptanalysis and

should not be used for protecting sensitive corporate communication today.